

An Immersed-Boundary Approach for Multilayer Ice Accretion Simulations

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Abstract

This work presents an immersed-boundary framework for aircraft ice-accretion simulations designed to reduce the dependence of conventional icing workflows on body-fitted volume meshes. The geometry is represented implicitly using a signed-distance level-set field and embedded within a non-conformal background mesh. A ghost-cell immersed-boundary treatment is used for the Reynolds-averaged Navier–Stokes flow solver, while both Eulerian and Lagrangian droplet formulations are extended to the immersed-boundary representation. Ice thermodynamics are evaluated using a Messinger-based formulation, and the resulting ice geometry is evolved using a continuous level-set method on a locally adapted mesh. The methodology is assessed using canonical droplet-impingement benchmarks and rime and glaze ice-accretion configurations on a NACA 0012 airfoil. Comparisons with body-fitted simulations demonstrate close agreement in the predicted collection-efficiency distributions and ice shapes. The results indicate that immersed-boundary methods provide a promising alternative for reducing the meshing constraints associated with multilayer aircraft ice-accretion simulations.

Introduction

Aircraft ice accretion is an unsteady multiphysics process involving aerodynamic flow, supercooled droplet transport, heat transfer, phase change, and the progressive evolution of the aircraft surface. Numerical icing tools generally exploit the difference between the characteristic time scales of these phenomena through a quasi-steady multilayer formulation. The total exposure time is divided into discrete accretion steps, during which the aerodynamic flow, droplet impingement, thermodynamic balance, and geometry evolution are solved sequentially.

Most numerical icing solvers adopt body-fitted computational meshes [1, 2]. The flow and droplet solutions are therefore obtained on a mesh conforming to the instantaneous aircraft geometry. Following each accretion step, the aerodynamic surface is updated and the surrounding volume mesh must consequently be deformed or regenerated before the next solution can be performed.

This requirement represents an important limitation for multilayer simulations. As the ice geometry becomes increasingly complex, repeated mesh deformation can deteriorate grid quality and eventually result in highly skewed or entangled elements. Although automated meshing approaches have progressed significantly [3], robust generation of high-quality meshes around complex three-dimensional ice features remains challenging, particularly for configurations involving highly irregular structures such as scallops [4].

Immersed-boundary methods offer an attractive alternative by embedding the geometry within a non-conformal background mesh. The computational grid is therefore no longer required to conform explicitly to the evolving surface. Instead, the presence of the solid boundary is introduced through modifications of the governing equations or reconstructed boundary values. Several immersed-boundary strategies have been investigated for icing, including penalization and ghost-cell formulations [5, 6, 7, 8, 9]. RANS-based immersed-boundary methodologies have also recently been applied within the AIAA Ice Prediction Workshop [1, 2, 10, 11].

The present work develops an immersed-boundary icing framework within the in-house CHAMPS solver. The approach combines a RANS immersed-boundary flow solver with Eulerian and Lagrangian droplet formulations, surface thermodynamics, and level-set-based geometry evolution. The objective is to reduce the dependency of multilayer ice-accretion simulations on body-fitted volume-mesh generation while retaining the predictive capability required for aerodynamic, impingement, and thermodynamic calculations.

Immersed-Boundary Ice-Accretion Framework

The complete numerical procedure is illustrated in Fig. 1. For each accretion layer, the aerodynamic flow field is first computed, followed by the droplet-impingement solution. The resulting collection efficiency is transferred to the thermodynamic model, which determines the local ice mass accretion rate. The geometry is then evolved using a continuous level-set method before the procedure is repeated for the subsequent layer.

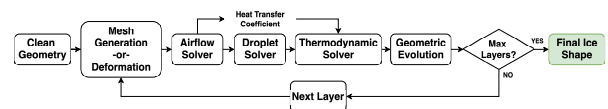


Figure 1: Multilayer ice-accretion framework combining immersed-boundary flow and droplet computations with level-set geometry evolution.

Immersed Geometry and Flow Solver

The immersed geometry is represented using a signed-distance level-set function ϕ . The solid region is identified by $\phi < 0$, the fluid region by $\phi > 0$, and the immersed surface corresponds to $\phi = 0$. The signed distance is obtained using a ray-casting procedure [12], while the gradient of the level-set field provides the local surface-normal direction,

$$\mathbf{n}_\phi = \frac{\nabla \phi}{|\nabla \phi|}. \quad (1)$$

Unlike a body-fitted formulation, the aerodynamic surface can intersect the background computational cells arbitrarily. Figure 2 illustrates the difference between the two representations for a circular cylinder.

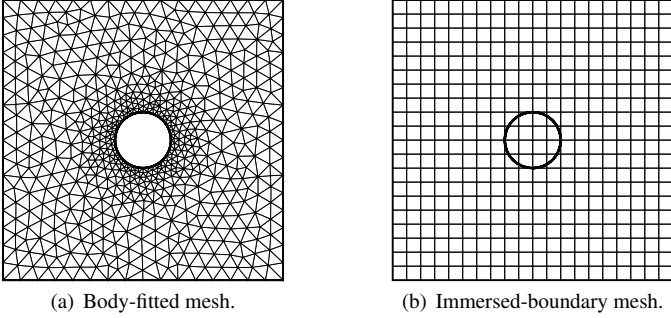


Figure 2: Body-fitted and immersed-boundary representations of the same geometry.

The aerodynamic solution is obtained from the compressible RANS equations, with turbulence modeled using the $k-\omega$ formulation [13]. Wall boundary conditions are imposed using a ghost-cell methodology based on [14]. Ghost cells located inside the immersed body are associated with image points in the fluid domain, from which the required flow quantities are reconstructed. This allows the boundary condition to be imposed sharply without modifying the connectivity of the background mesh.

Eulerian and Lagrangian Droplet Impingement

Both Eulerian and Lagrangian descriptions of the dispersed water phase are considered. Under typical aircraft icing conditions, the droplet phase is sufficiently dilute for droplet–droplet interactions to be neglected [15].

In the Eulerian formulation, the droplet conservation equations are solved directly on the background mesh. The presence of the immersed solid is introduced through a penalization procedure following the formulation developed in [7]. This strategy requires only limited modification of the original governing equations and allows the same non-conformal computational mesh to be used for both aerodynamic and droplet solutions.

The Lagrangian formulation instead computes individual droplet trajectories through the aerodynamic velocity field. Trajectories are advanced through the unstructured background mesh following the tracking procedure described in [16]. Intersection between a droplet trajectory and the immersed level-set surface identifies an impact event and provides the information required to reconstruct the local collection efficiency.

The availability of both formulations provides an additional verification of the immersed-boundary impingement methodology. The Lagrangian formulation also avoids some of the numerical diffusion associated with Eulerian transport near sharp impingement limits.

For supercooled large droplets (SLD), additional deformation and splashing effects are incorporated. Droplet drag is corrected following [17], while the semi-empirical splashing formulation of [18] accounts for the influence of impact angle and kinetic energy.

Thermodynamics and Geometry Evolution

Surface thermodynamics are evaluated using a Messinger-type formulation following [19]. The local energy balance determines the fraction of impinging water that freezes and provides the corresponding ice mass accretion rate.

The resulting geometry is evolved using a continuous level-set method based on the formulation of [20]. The ice–air interface is represented by the zero contour of ϕ and propagated according to

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{V}) - \phi \nabla \cdot \mathbf{V} = 0, \quad (2)$$

where $\mathbf{V}$ represents the local ice-growth velocity. The magnitude of this velocity is determined from the predicted ice mass accretion rate and ice density.

Although the aerodynamic and droplet calculations are performed using the immersed-boundary background mesh, a hybrid strategy is currently adopted for geometry evolution. The level-set equation is solved on a locally adapted unstructured mesh refined around the ice-growth region, as shown in Fig. 3.

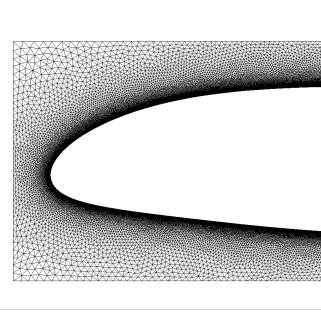


Figure 3: Locally adapted unstructured mesh used for level-set geometry evolution around the iced surface.

A semi-implicit Rosenbrock–Wanner integration scheme is employed to reduce the restrictive time-step requirements associated with explicit level-set propagation. Following each accretion step, the zero level set is extracted using the MMG remeshing library [21], producing the updated ice surface for the following layer.

Preliminary Results

The immersed-boundary framework is assessed using canonical droplet impingement cases followed by complete rime and glaze ice-accretion simulations. A circular cylinder is first considered because an analytical collection-efficiency solution is available for verification [7]. Both the Eulerian and Lagrangian immersed-boundary formulations recover the expected collection-efficiency distribution and show close agreement with their corresponding body-fitted solutions.

The methodology is further assessed using the NACA 23012 SLD impingement benchmark of [22]. Figure 4 compares the predicted collection-efficiency distributions obtained with the Eulerian and Lagrangian formulations. Inclusion of SLD deformation and splashing effects improves agreement with the experimental measurements. Both immersed-boundary formulations reproduce the overall impingement behavior, although the Eulerian solution exhibits slightly greater numerical smearing near sharp impingement limits, consistent with previous observations for Eulerian immersed droplet formulations [7].

Complete ice-accretion simulations are subsequently performed for rime and glaze conditions on a NACA 0012 airfoil using the wind-

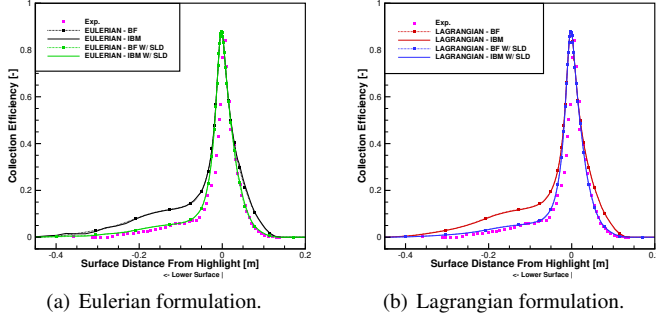


Figure 4: Collection-efficiency distributions for the NACA 23012 SLD impingement benchmark of [22].

tunnel experiments of [23]. For the rime condition, where the ice geometry is predominantly controlled by the impingement distribution, the body-fitted and immersed-boundary predictions are nearly identical. Increasing the number of accretion layers also improves agreement with the experimental profile.

The glaze condition provides a more demanding assessment because the resulting horn geometry depends strongly on the local thermodynamic balance and convective heat transfer. Figure 5 compares body-fitted and immersed-boundary predictions after one and five accretion layers. Both approaches reproduce the principal experimental features, including the formation of the upper and lower horns. The single-layer predictions remain close, whereas differences become more visible after five layers. These deviations are primarily associated with differences in the near-wall heat-transfer prediction, which accumulate as the geometry evolves.

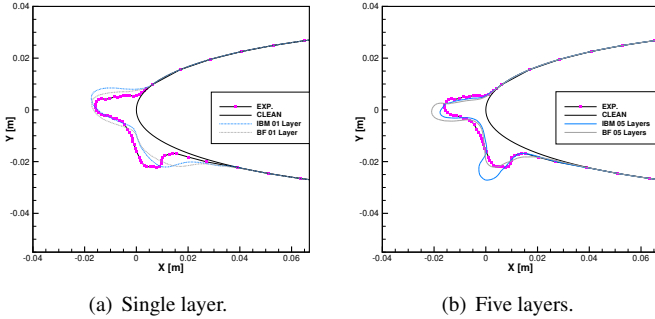


Figure 5: Body-fitted and immersed-boundary ice-shape predictions for the NACA 0012 glaze-ice configuration of [23].

Conclusion and Future Work

An immersed-boundary framework for multilayer aircraft ice accretion has been developed within CHAMPS. The methodology combines a RANS ghost-cell immersed-boundary flow solver, Eulerian and Lagrangian droplet transport formulations, a Messinger-based thermodynamic model, and level-set-based geometry evolution. The approach reduces the dependence of the aerodynamic and droplet calculations on body-fitted volume meshes while retaining a sharp representation of the immersed surface.

Verification using a canonical cylinder demonstrates that both Eulerian and Lagrangian immersed-boundary droplet formulations recover the expected collection-efficiency distribution. Application to the NACA 23012 SLD benchmark further demonstrates good agreement with experimental impingement measurements, while the Lagrangian formulation provides a sharper representation of the impingement limits. Complete rime and glaze ice-accretion simulations on a

NACA 0012 airfoil show that the immersed-boundary approach produces ice shapes comparable to those obtained using the body-fitted formulation and captures the principal experimental features.

The results also identify the aerodynamic and thermal wall treatment as an important area for further development. While differences between body-fitted and immersed-boundary solutions remain limited for rime conditions, small variations in convective heat transfer can accumulate over successive layers under glaze conditions and influence the final horn geometry.

Future work will therefore focus on improving the immersed-boundary near-wall treatment for aerodynamic and thermal predictions and on extending the methodology to fully three-dimensional ice-accretion configurations. The longer-term objective is to perform the complete multilayer icing workflow on non-conformal meshes, further reducing the need for repeated body-fitted mesh generation as increasingly complex three-dimensional ice features develop.

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Level-set interface normal
Time
Ice-growth velocity

$[-]$
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 $[m/s]$

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Acknowledgements

This work was supported in part by the Canada Research Chairs Program, the Natural Sciences and Engineering Research Council of Canada (NSERC), and the Digital Research Alliance of Canada.

Definitions, Acronyms, Abbreviations

CFD	Computational Fluid Dynamics
IBM	Immersed-Boundary Method
RANS	Reynolds-Averaged Navier–Stokes
SLD	Supercooled Large Droplets

General symbols

Greek symbols

ϕ

Level-set signed-distance function

$[m]$